

Mathematical and Logical Foundations of Computer Science

Theorem proving

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(some slides were adapted from Benedikt Ahrens' slides)

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Where are we?

- ▶ Symbolic logic
- ▶ Propositional logic
- ▶ Predicate logic
- ▶ **Intuitionistic vs. Classical logic**
- ▶ **Type theory**

Today

Theorem proving using Lean

- ▶ Propositional Logic in Lean
- ▶ Proofs as programs in Lean

Useful links:

- ▶ online version:
<https://leanprover.github.io/live/master/>
- ▶ reference manual:
<https://leanprover.github.io/reference/index.html>
- ▶ theorem proving in Lean: https://leanprover.github.io/theorem_proving_in_lean/index.html

Further reading:

- ▶ Chapter 4 of
http://leanprover.github.io/logic_and_proof/

Theorem Proving - What and Why?

Theorem provers (or **proof assistants**) are formal verification tools that help establish claims

Of both

- ▶ **mathematical results**
- ▶ **hardware & software verification**

Why use formal verification tools?

- ▶ computer checked proofs are **less error prone** than pen-and-paper proofs
 - ▶ those tools can check that we do not make mistakes
 - ▶ they can handle low level details “fairly” automatically
 - ▶ they can easily check and re-check proofs
- ▶ To **rely on safety-critical systems**, they need to be checked/verified to the highest standards possible
 - ▶ Such as power plants, airplanes, autonomous vehicles, etc.
 - ▶ failures can be catastrophic (loss of lives, money, data, etc.)!

Theorem Proving - What and Why?

Example of a mistake in Mathematical proofs: Voevodsky (Fields medalist) discovered a mistake in one of his paper. *“A technical argument by a trusted author, which is hard to check and looks similar to arguments known to be correct, is hardly ever checked in detail”* - Vladimir Voevodsky, 2014,

<https://www.ias.edu/ideas/2014/voevodsky-origins>

Many examples of costly hardware & software failures:

- ▶ 1994: Intel Pentium bug in the floating point unit
- ▶ 1996: ESA (European Space Agency) Ariane 5 launcher self-destructs
- ▶ 2010/2015: Toyota Prius software “glitches”
- ▶ 2012/2014: Heartbleed security bug in the OpenSSL crypto library
- ▶ 2018: TSB online banking “glitches”

Could we avoid these issues using formal verification?

Theorem Proving - Automated vs. Interactive

Several kinds of formal verification tools:

- ▶ **automated**

- ▶ e.g., SAT/SMT solvers such as Z3
- ▶ e.g., Model checkers such as Spin, Prism
- ▶ automatically **search** solutions to problems

- ▶ **interactive**

- ▶ e.g., Agda, Coq, Lean, etc.
 - ▶ provide ways of **building** and **verifying** proofs
 - ▶ every single proof step is checked to be correct w.r.t. some logical foundations
- ▶ many systems, such as Lean, support both automated and interactive theorem proving

Theorem Proving - Formal Verification

Formal verification tools such as theorem provers are especially useful in computer science to

- ▶ prove **mathematical** results, e.g.,
 - ▶ four-color theorem
 - ▶ odd-order theorem
- ▶ prove the correctness of **hardware & software systems**, e.g.,
 - ▶ CompCert: a C compiler verified in Coq
 - ▶ CakeML: a verified ML language in HOL4
 - ▶ sel4: a microkernel in Isabelle

Many companies nowadays use formal verification tools to ensure the correctness of their hardware & software systems, such as (see <https://github.com/ligurio/practical-fm>):

Airbus	Amazon	ARM
BAE Systems	Data61	Ethereum
Facebook	Galois	Google
Microsoft	NASA	...

Theorem Proving - Formal Verification

In hardware & software system verification, formal verification tools are used to **mathematically prove** that **systems** (programs or models) satisfy formal **specifications**

Example: For a sorting algorithm, we want to prove that the result is sorted

Example: For a “remove” algorithm that removes an element from a collection, we want to prove that the resulting collection does not contain the element

Theorem Proving - Formal Verification

Benefits and drawbacks of formal verification by theorem proving:

- ▶ exhaustive (cover all cases)
- ▶ rigorous (includes all required logical steps)
- ▶ difficult and time consuming
- ▶ lead to large proofs
- ▶ relies on the correctness of the prover

Theorem Proving - Formal Verification

We will now show how to prove mechanically-checked theorems using the **Lean** theorem prover

- ▶ developed by Microsoft Research
- ▶ it runs in the browser
- ▶ you can install it on your machines

Lean implements a logical system:

- ▶ it is based on a **dependent type theory**, which includes predicate and propositional logic
- ▶ it supports **constructive reasoning** and has a **computational interpretation** that allows viewing **proofs as programs** and **propositions as types**
- ▶ it implements a sequent calculus
- ▶ it relies on **tactics** to apply rules (we will see a few today)

Theorem Proving in Lean

Let us switch to Lean now

<https://leanprover.github.io/live/master/>

see the file called `prop.lean`

Conclusion

What did we cover today?

- ▶ Theorem proving
- ▶ Propositional Logic in Lean
- ▶ Proofs as programs in Lean

Further reading:

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Next time?

- ▶ Theorem proving